

Collision Frequency Trends for Various Vehicle Types in the Greater Toronto Area*

INF2167 Project - Group 7

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Provide a short (~150 word) abstract that includes the main problem you investigated, your data and methods, findings, and contribution(s).

*Code and data are available at: https://github.com/srg-uoft/Toronto_Vehicle_Collision_Frequency_Project.

1 Introduction

1.1 Project Overview

a. Provide an overview of your project, the data and methods, and main finding(s). we can probably re-use some things from the proposal introduction section for this, but should make sure we modify it to discuss our methods and findings.

1.2 Research Questions

b. Clearly state the research question(s) you investigated.

1.2.1 What are the most accident-prone times for travelers in the GTA, and how does that vary depending on mode of transportation?

The processed dataset allows us to compare risk profiles across modes, thanks to a indicator for involvements of different modes of travel.

1.2.2 Does the GTA's data align with pre-existing research on riskiest travel times?

Prior research frequently identifies holidays as high risk periods for collisions. By analyzing temporal patterns in the TPS dataset, we can test whether Toronto follows these broader trends or shows unique local patterns. This comparison helps contextualize Toronto's traffic safety landscape within the wider literature on collision risk.

1.3 Hypotheses

c. Include any hypotheses and explain why these questions are of interest. this will be important, since a feedback point we got on our proposal was that we needed to more clearly state our hypotheses

2 Relevant Literature

Scholars consistently conclude that road traffic collisions cluster in both location and time, thus we are inspired on discovering when and where risk is highest in the GTA. A spatial analysis of road traffic injuries is conducted in the Greater Toronto Area (Vaz et al. 2017) using Moran's I and Getis Ord Gi. The article concludes that collisions form statistically significant clusters rather than occurring randomly. Their results show that "Peel and Durham have the highest collision rate", which reminds us of the importance to identify hotspots of collisions in different neiboughoods. This directly provides us preview on Toronto collisions by communities. Temporal patterns in collision risk are also documented already by scholars. An article compares crashes during statutory holidays and regular weekends in Alberta (Anowar et al. 2013). They find that severe collision occurrences

are less than estimated. Certain days (especially major holidays) carry elevated collision risk, which inspires our interest in comparing GTA collision patterns to commonly recognized high risk days.

3 Dataset & Methods

3.1 Data Source

The Toronto Police Service (TPS) maintains a Public Safety Data Portal, where they regularly publish and update crime statistics datasets for the Greater Toronto Area. The data used in this analysis is the Traffic Collisions Open Data (ASR-T-TBL-001) dataset, obtained from (Toronto Police Service 2023) and licensed under the Open Government Licence – Ontario. A notable limitation is that only collisions where TPS were called to the scene are included in this dataset - if for any reason, the involved parties made a decision to not involve the police, their collision statistics will not be reported here.

3.2 Data Structure & Key Variables

Per (Toronto Police Service 2023), each row/observation in this dataset represents a single collision, and the columns/variables use identification variables to describe each collision in three ways: 1) where and when the collision occurred, 2) whether there were injuries, deaths, or property damage, and whether parties failed to remain at the scene, and 3) which types of vehicles were involved in the collision. To answer the question of where the highest risk of traffic collision for various modes of transportation is in the Greater Toronto Area, our primary variables of interest are the spatial and temporal variables (to determine where and when the most accidents occur), and the information about involved vehicles (to determine any diverging trends between modes of transportation).

3.3 Data Preparation: Cleaning & Pre-Processing

The raw data as obtained from (Toronto Police Service 2023) was cleaned in a variety of ways before analysis. This cleaning was done using the programming language R (R Core Team 2026) due to its powerful statistical and data-focused capabilities. To improve readability and minimize errors, several columns were renamed using more semantic language. As an example, the column “FTR_COLLISIONS,” which contains information about whether one or more parties failed to remain at the collision site, was renamed to “FAILURE_TO_REMAIN,” so this meaning is more plainly understandable. After renaming select columns, all columns were standardized to snake_case using the janitor library (Firke et al. 2024) to further improve consistency and cleanliness in the working data. Finally, two new columns were added to make statistical visualizations easier to generate for ordered temporal variables: a column named “occ_month_num” to represent the month names for each collision as numbers from 1-12 (January = 1, June = 6, etc.), and a similar column named “occ_month_dow” to represent the days of the week for each collision as numbers from 1-7 (Sunday = 1, Wednesday = 4, etc.).

3.4 Exploratory Data Analysis

Initial exploratory data analysis (EDA) was done in R (R Core Team 2026) making heavy use of the Tidyverse library (Wickham et al. 2019), with the goal of uncovering high-level trends and possible variables of interest for further analysis. To improve reproducibility, all filepaths were made relative using the Here package (Müller 2025), and project contents were regularly committed to GitHub for granular version control.

The first EDA step was simply looking at the temporal and spatial data, to make sure that there were no obvious encoding errors, and that the values in our key variable columns made sense. This was done using R’s `unique()` function on the columns `occ_hour` (hour the collision occurred), `occ_dow` (day of the week the collision occurred), `occ_month` (month the collision occurred), and `neighbourhood` (Toronto neighbourhood where the collision occurred). While all the values in the temporal columns were encoded consistently and correctly, this did reveal that the `neighbourhood` column had an “NSA” value, representing a collision where no location was recorded. Using `group_by()` and `mutate()` to calculate the percentage of total collisions which occurred in each neighbourhood revealed that 17% of collisions did not have location data recorded. While this does not make reporting or statistical modelling impossible for spatial data, this is a limitation of the data, and it is worth noting that our conclusions may not be as reflective of our target population as they would have without this missing data.

Since the second part of our research question involves how spatial and temporal trends in collision frequency might change depending on mode of transportation, it was also important to look at the vehicle data available, to understand how it might be used and make sure there were no encoding errors. In the original dataframe, the five vehicle types (*automobile*, *bicycle*, *motorcycle*, *passenger (public transit)*, and *pedestrian*) are reported as separate columns. Using the `unique()` function again revealed that within each of the 5 vehicle type columns, the value of each cell contains either “YES” or “NO” to indicate whether a vehicle of that type was involved in that collision. To better analyze vehicles that were actually involved in crashes, and also to adhere better to Tidy data principles (Wickham 2014), R’s `pivot_longer()` function was used to melt the vehicle type columns from the original dataframe into one column containing vehicle type (`vehicle_type`) and another containing the binary cell values (`involved_yesno`), thereby changing the observational unit from individual collisions to individual vehicles involved in collisions with five rows per collision. This long dataframe was then filtered to only vehicles marked “YES” so as to visualize and model vehicles involved in collisions.

Understanding the key variables to answering our research question, statistical visuals, and summary tables were generated using the *ggplot2* library from (Wickham et al. 2019) and (Zhu [aut et al. 2026]). Selections from these that best contribute to answering the research question, as well as discussions on the limitations and implications of the trends revealed by these visuals, can be found in the later sections of this paper.

3.5 Statistical Modeling - Linear Regression

To better understand the relationship between collision frequency and the key temporal, spatial, and vehicle type variables described above, the final method employed in this paper will be linear

regression modelling, using the key variables as predictors and number of collisions as an outcome variable. The model will take the form of the following notation:

$$y = \beta_0 + \beta_1 x_1 + \dots + \beta_n x_n + \varepsilon$$

where y is number of collisions (the outcome variable we are looking to predict), β_0 is the intercept (the number of collisions when all predictors equal zero), β_n represents the change in number of collisions for a one-unit increase in the n predictor, x_n is the predictor in question, and ε represents the residual error present in the model. This will be translated into R model syntax using the following code:

```
model <- lm( ~ x_1 + ... + x_n, data = tps_collisions)
```

where y represents the number of collisions for a given combination of predictors, and x_1 through x_n are the used predictors.

Generated models will use built-in R modelling functionality (R Core Team 2026), visualization capabilities from the broom package (Robinson et al. 2026), and summary functions from the modelsummary package (Arel-Bundock 2022). The performance of any generated models will be reported and discussed further in the sections below.

4 Results & Findings

Overall, when looking at spatial and temporal data for vehicles involved in collisions, we see similar trends across vehicle type for collision hour and location, but notably diverging trends for collision day-of-week and collision month. Summary statistics are presented in Figure 1, and temporal trends are elaborated on in more depth below.

Vehicle Type	Hour with Highest Number of Recorded Collisions	Day of Week with Highest Number of Recorded Collisions	Month with Highest Number of Recorded Collisions	Neighbourhood with Highest Number of Recorded Collisions
Automobile	15	Friday	February	Wexford/Maryvale (119)
Bicycle	17	Wednesday	July	Annex (95)
Motorcycle	17	Saturday	August	Wexford/Maryvale (119)
Public Transit Vehicle	17	Saturday	January	Wexford/Maryvale (119)
Pedestrian	18	Wednesday	November	Wexford/Maryvale (119)

Figure 1: Hour, Day-of-Week, Month, and Neighbourhood with the Most Collisions for All Vehicle Types

4.1 Afternoon Peak for All Vehicle Types

Looking at Figure 2, we see that trends in collision frequency are similar across all five vehicle types. Overnight collisions (21:00 - 04:00) are low, before a sharp increase between 05:00-09:00. We then see a steady increase throughout late morning and afternoon, a daily high between 15:00 and 18:00, and then a steady decline through the evening for all vehicle types. While collisions frequency for automobiles is substantially higher than for other vehicle types (looking at the differing y-axis scales for the various vehicle plots), the daily pattern remains the same.

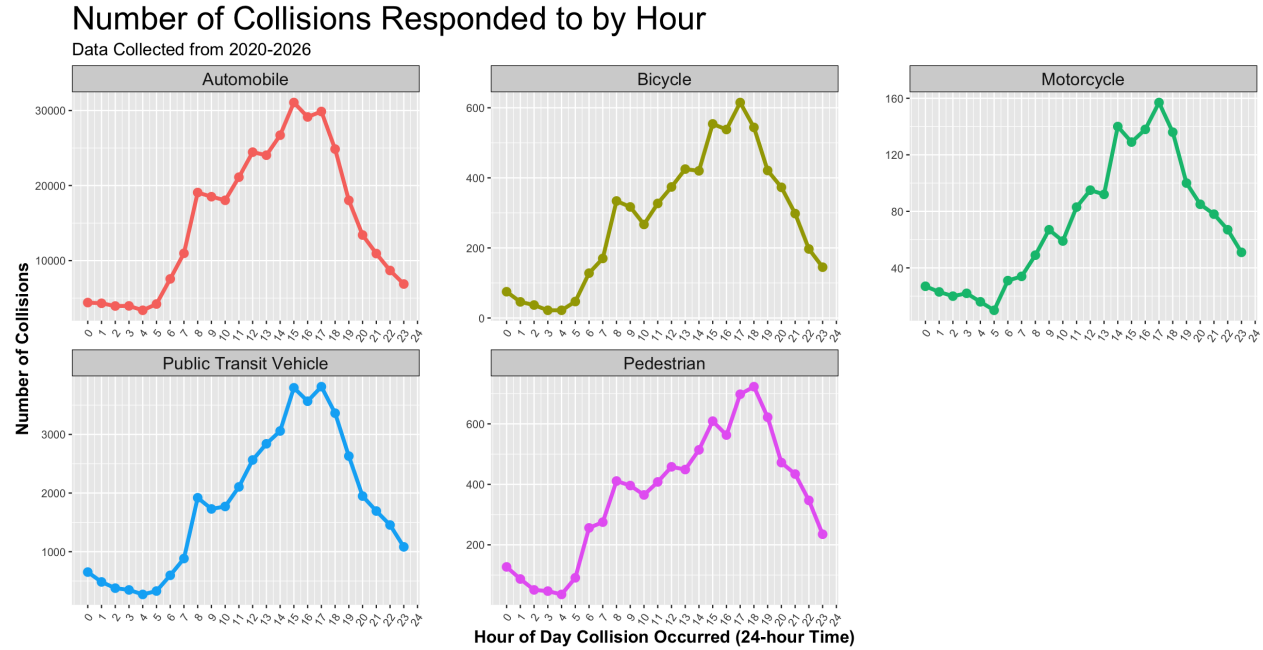


Figure 2: Collisions Per Hour are Fairly Consistent Across Vehicle Types

4.2 Weekday vs Weekend Trends

Zooming out to look at trends per day of the week in Figure 3, there are two primary groups. Automobiles, bicycles, and pedestrians have low collision frequencies on weekend days (Saturday and Sunday) and higher numbers during the week, while motorcycles and public transit vehicles have more collisions clustered towards the end of the week and the weekend. Bicycles and pedestrians experience weekly peak collisions in the middle of the week (Tuesday - Thursday), while automobiles experience a steady increase throughout the week, a sharp peak on Friday, and then a sharp decrease during the weekend. Friday and Saturday are peak days for both motorcycles and public transit vehicles.

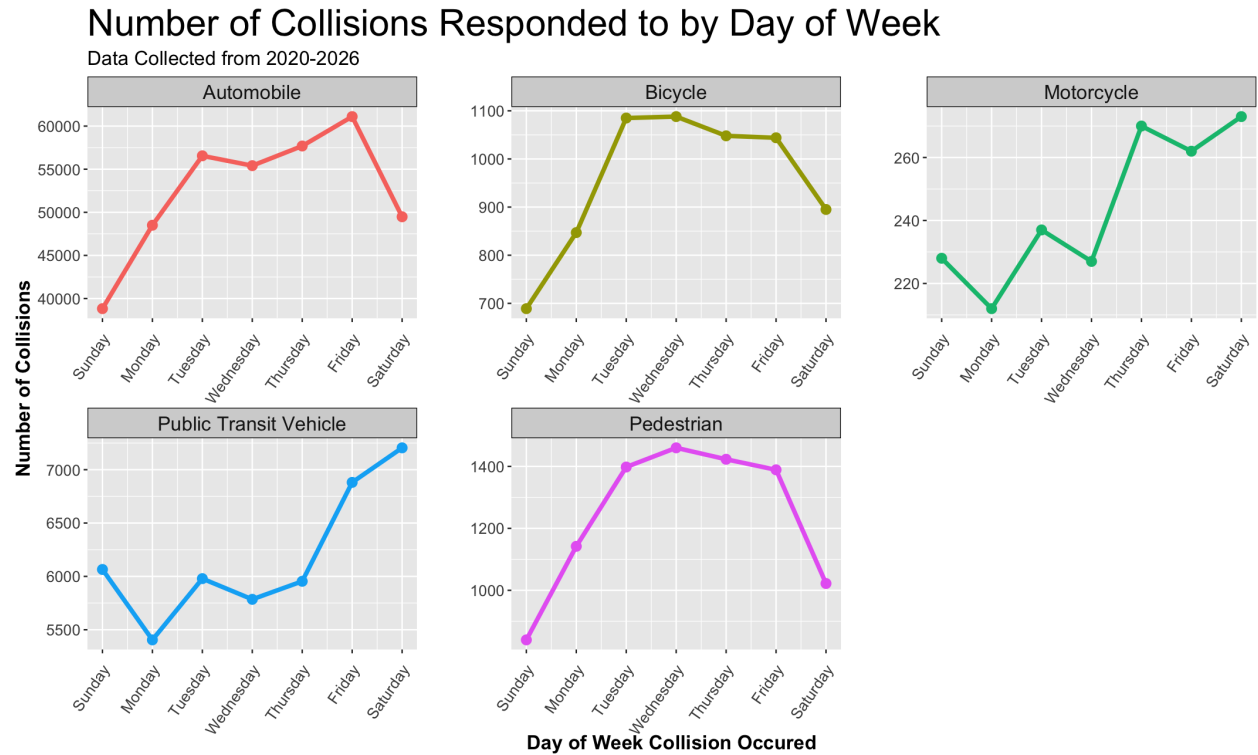


Figure 3: Automobile, Pedestrian, and Bicycles Have More Weekday Collisions, while Motorcycles and Public Transit Vehicles Have More Weekend Collisions

4.3 Opposing Seasonal Trends

Considering collisions per month across the year in Figure 4, similar to looking at day-of-week trends, two main patterns emerge. For bicycles and motorcycles, collisions sharply increase between April and May, stay high through September, and then drop sharply through the end of the year. However, for automobiles, pedestrians, and public transit vehicles, the opposite trend is true: collisions are at a yearly high towards the beginning of the year (for automobiles and public transit vehicles) or the end of the year (for pedestrians), see a sharp drop around April, and then steadily climb again to the end of the year.

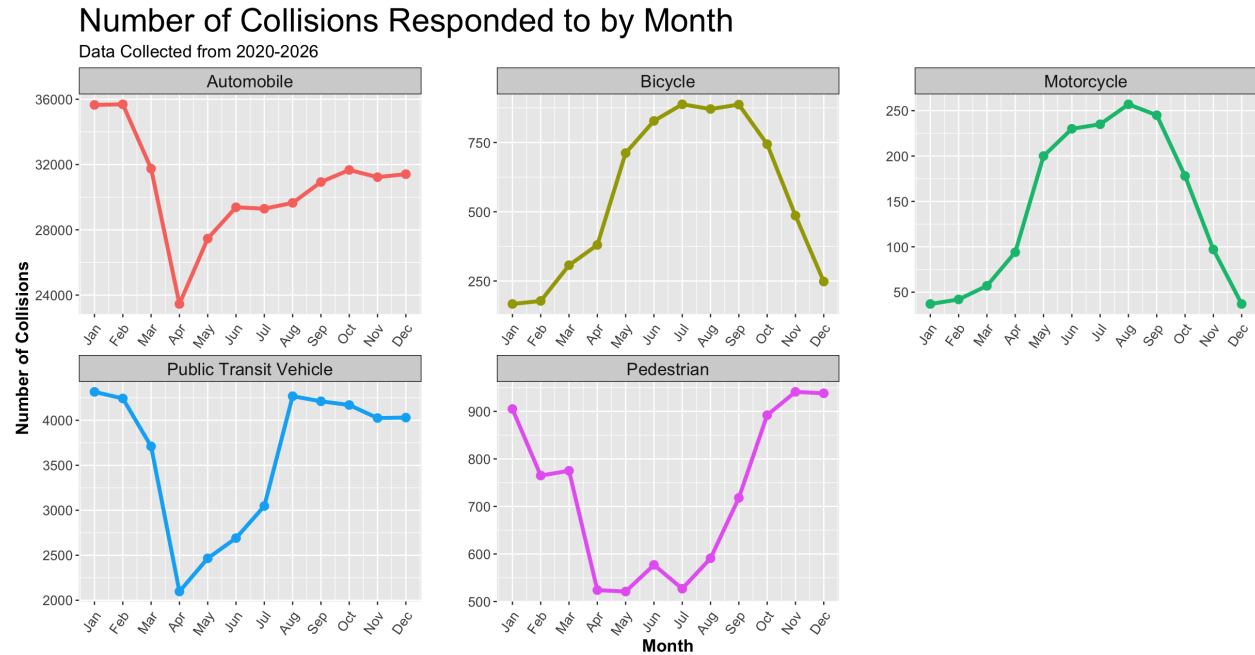


Figure 4: Bicycles and Motorcycles Experience Yearly Highs from April - August, While Automobiles/ Public Transit Vehicles, and Pedestrians Experience Yearly Lows

4.4 Modeling Spatial, Temporal, and Vehicle Type Relationships

As mentioned above, linear regression models were run to better understand the relationships between the key variables to answering our research question. The first model, `overall_model`, was run using neighbourhood, hour, day-of-week, month, and vehicle type as predictors, attempting to predict the number of collisions that will happen in any combination of the predictor variables. The model as written in R syntax is:

```
overall_model <- lm(number_of_collisions ~ neighbourhood + occ_month + occ_dow +
  occ_hour + vehicle_type, data = overall_collisions)
```

The R^2 value for this model is **0.3241**.

The second model run, `minus_vt_model`, was run to investigate if model performance would improve if vehicle type were removed as a predictor. Since in the EDA phase we determined that a vast majority of vehicle data points in the long data frame were automobiles, having such a disproportionate predictor might be skewing the model. The second model as written in R syntax is:

```
minus_vt_model <- lm(number_of_collisions ~ neighbourhood + occ_month + occ_dow
  + occ_hour, data = overall_collisions_minus_vt)
```

The R^2 value for this model is **0.5806**.

To evaluate the models and how well they fit the data side-by-side, Figure 5 plots the residual errors from both models versus their respective predictions.

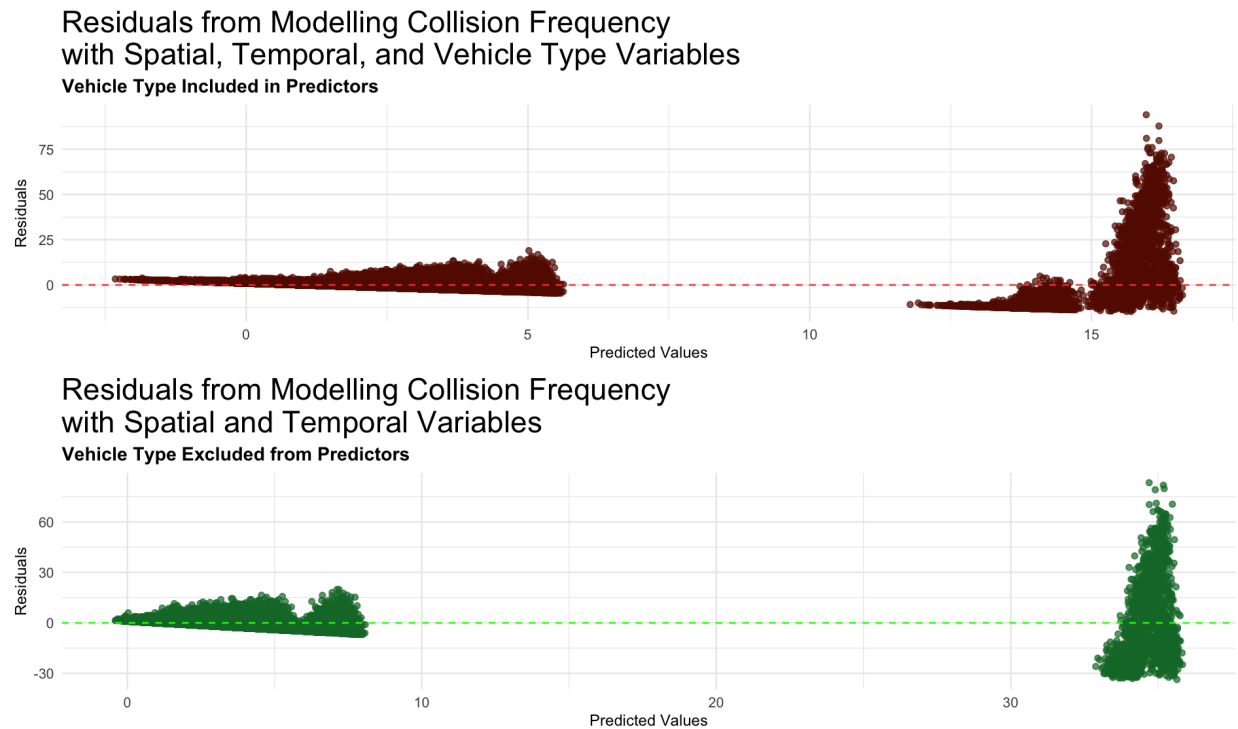


Figure 5: Residual Plots for Two Models, One Including Vehicle Type as a Predictor and One Excluding Vehicle Type

5 Discussion

5.1 Findings & Implications

5.2 Limitations

5.3 Future Research

6 Conclusion

7 References

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